# Fair Use and AI-Copyrightability Must Be Designed Jointly

## Synthesizing Yang & Zhang (2024) and Cooper et al. (2025)

Audit synthesis

Yang & Zhang (2024), *arXiv:2402.17801v2* | Cooper, Martens, Peukert, & Stocker (2025), *Review of Network Economics* 24(3): 161–176

# $1.5 billion

the largest copyright recovery in history

paid by one AI lab — for downloads, not training.

And the doctrine is still unsettled.

Bartz v. Anthropic; Cooper et al. 2025, p. 165

# GenAI breaks copyright at both input and output stages

| Stage | Core question | Status |
|---|---|---|
| **Input** | Can copyrighted works be scraped for training? | Unsettled: fair use (US), TDM & opt-outs (EU) |
| **Output** | Are AI-assisted works copyrightable? | Disparate thresholds: US, EU, China |

Cooper et al. (2025), Sections 3.1–3.3.

# Two-thirds of AI training datasets carry wrong or missing licenses

- $\sim$**65%** of datasets on Hugging Face & PapersWithCode have missing, incorrect, or ambiguous license metadata.
  (Longpre et al. 2024a, cited in Cooper 2025, p. 167)
- In **LAION-5B** (largest public image set): fewer than **1 in 1,000** datapoints carry license metadata.
- **10 website domains** account for **23%** of LAION's $\sim$2 billion images — supply is concentrated, not representative.
- Opted-out vs. licensed images depict semantically different concepts and differ in technical quality.
  (Shcherbakov et al. 2025, cited in Cooper 2025, pp. 167, 171)

# The two leading US rulings disagree on why training is fair use

| | |
|---|---|
| **Alsup (Anthropic)** | Initial book downloads infringing $\rightarrow$ **$1.5B** recovery; subsequent training use ruled fair use. |
| **Chhabria (Meta)** | Training fair use *only because* plaintiffs "made the wrong arguments"; judge believes training would generally be illegal. |
| **USCO (2025b)** | Doubts fair use applies when AI outputs compete in existing markets for training works. |
| **EU — Hamburg** | LAION ruling: terms-of-service opt-outs likely enforceable. |

Cooper et al. (2025), pp. 165–166. Dozens of US suits remain pending.

# The dynamic problem

What is welfare-maximizing today can be welfare-destroying tomorrow.

# When training data is abundant, generous fair use maximizes *every* welfare metric

| **Welfare variable** | **Maximized by** |
|---|---|
| Model quality $X_2$ | $f = 0 \;\rightarrow\; X_2 = 1$ |
| AI content quantity | $f = 0$ |
| Creator income | $f = 0$ |
| Consumer surplus | $f = 0$ (all tested $X_1, k, M$) |
| Social welfare | $f = 0$ |

**Optimal joint policy:** $\{f = 0,\ \phi = 1\}$ — generous fair use *plus* full AI-copyrightability.

Yang & Zhang (2024), Proposition 7 & Corollary 2; data-abundant regime, $Q_0 \rightarrow \infty$.

# When training data is scarce, generous fair use harms AI development

- **Generous fair use** ($f \to 0$): kills creator incentives $\to$ no human content for training $\to$ model improvement stalls. *(Prop. 10)*
- **Strict fair use** ($f$ too large): AI company abandons model improvement entirely. *(Prop. 11)*
- **A moderate $f$ maximizes** model quality $X_2$, social welfare, and consumer surplus.

**The central novel finding**

A copyright doctrine welfare-optimal in 2020 (data-abundant) becomes welfare-destroying once pre-existing data is exhausted (data-scarce, $Q_0 = 0$).

Yang & Zhang (2024), Propositions 10–11.

# Strong AI-copyrightability flips fair use from helpful to harmful

| AI-copyrightability $\phi$ | Effect of generous fair use on AI development |
|---|---|
| **Weak** | **Promotes** development<br>creators still produce human content for training |
| **Strong** | **Hinders** development<br>too few creators $\rightarrow$ training data dries up |

$X_2$ is inverted-U in $\phi$ when data is scarce *(Prop. 12)*. The two policies cannot be designed independently.

Yang & Zhang (2024), Figure 9 & Proposition 12.

# The output side

Three jurisdictions, three verdicts, one work.

# Three jurisdictions reach three verdicts on the same AI-assisted work

| Jurisdiction | Ruling | Threshold |
|---|---|---|
| **China** *(Li v. Liu, 2023)* | **Copyrightable** | “Intellectual investment” & “personalized expression” via prompting |
| **US** *(USCO 2025a)* | **Not copyrightable** | “All current-state prompting outputs” lack sufficient human control |
| **EU** *(Czech)* | **Not copyrightable** | “Plaintiff did not personally create the work” |

Cooper et al. (2025), Section 3.2, pp. 168–170.

# Disclosure cannot enforce the AI / non-AI line — and creates perverse incentives

- **The "Old Hard Drive Debacle"** *(Cooper 2025)*: artists must log every GenAI use solely to strip themselves of rights → strong incentive to *deny* AI use.
- **No robust detection tool** exists for whether — let alone *which elements* of — a work used GenAI.

  (10+ sources on watermarking limits cited in Cooper 2025)
- **Convergence problem:** search-engine and AI-training crawlers are technologically indistinguishable; blocking one blocks the other.

Cooper et al. (2025), Section 3.2, pp. 169–170, 172.

# Synthetic data is no escape — model collapse and copy-mining await

- GenAI produces near-identical copies of protected works even with developer safeguards.
  (Cooper & Grimmelmann 2025)
- US "Blurred Lines" precedent: weak similarity thresholds mean outputs need not be close replicas to infringe.
- **Copy-mining incentive**: mass-generate to claim maximum rights, mining out the latent space between existing works.
  (Cooper 2025; Lemley 2024)
- **Synthetic data fails as a fix:** depends on real-data quality, cannot generate genuine novelty, recursive training risks model collapse.
  (Shumailov et al. 2024)

# Optimal policy depends on objective *and* data regime — divergence is rational

| Regulator's objective | Optimal $\phi$ | Optimal $f$ |
|---|---|---|
| Consumer surplus | Lowest | More generous |
| Social welfare / AI development | Substantially higher | Stricter |

- Fair use is more sensitive than $\phi$ to data availability ($Q_0$), training efficiency ($k$), and market growth ($M$).
- Predicted alignment: **US / China** $\rightarrow$ AI growth; **EU** $\rightarrow$ consumer welfare $\rightarrow$ persistent jurisdictional divergence.

Yang & Zhang (2024), Figure 10; Cooper et al. (2025), citing Peukert 2025a.

# A skeptic would say the model is too stylized to drive real policy

**The objection.** Yang & Zhang assume uniform skill, competitive AI markets, binary content quality, and a single regulator. Cooper et al. provide no original empirical evidence.

**The response.**

- Yang & Zhang's regime-dependence result (abundant vs. scarce) is robust across all numerical examples and does not hinge on the uniform-skill closed form.
- The Cooper primer's job is not estimation — it identifies empirical gaps (representativeness of opt-out data, creator-behavior responses, output memorization rates) for the next generation of work.
- Both papers converge on the same conclusion from independent methods: copyright doctrine cannot be optimized one lever at a time, in one jurisdiction, at one moment in time.

# Joint. Dynamic. Cross-border.

Fair use and AI-copyrightability are economically linked.

Designing them in isolation destroys welfare once data runs scarce.